

Study notes

DEFINITION • `math:algebra:df:group`

Group

Recall target. State all four parts and distinguish a group from a monoid.

A **group** is a pair $(G, \cdot)$ consisting of a set G and a binary operation $\cdot : G \times G \rightarrow G$ such that:

1. **Associativity:** $(ab)c = a(bc)$ for all $a, b, c \in G$.
2. **Identity:** there is an element $e \in G$ such that $ea = ae = a$ for every $a \in G$.
3. **Inverses:** for each $a \in G$, there is an $a^{-1} \in G$ such that $aa^{-1} = a^{-1}a = e$.

Closure is encoded by requiring the operation to have codomain G .

Alternative: One-object category

A **group** is a category with one object in which every morphism is invertible.

If the unique object is $*$, the group elements are the morphisms $\mathbf{Hom}(*, *)$, multiplication is composition, the identity morphism is the group identity, and categorical inverses are group inverses.

REMARK • `math:algebra:rk:associativity-not-commutativity`

Associativity is not commutativity

Associativity controls **parenthesization**:

$$(ab)c = a(bc).$$

Commutativity controls **order**:

$$ab = ba.$$

Neither property implies the other. Matrix multiplication is associative but is generally not commutative.

Lagrange's theorem

Dependencies. Finite groups, subgroups, left cosets, and the fact that cosets partition a group.

Let G be a finite group and let $H \leq G$. Then

$$|G| = [G : H] |H|,$$

where $[G : H]$ is the number of left cosets of H in G .

Alternative: Divisibility form

If H is a subgroup of a finite group G , then $|H|$ divides $|G|$.

In particular, the order of every element $g \in G$ divides $|G|$, because it is the order of the cyclic subgroup $\langle g \rangle$.

The converse is false in general: a divisor of $|G|$ need not occur as the order of a subgroup.

Proof

The left cosets of H partition G . For each $g \in G$, left multiplication

$$h \mapsto gh$$

is a bijection from H to gH , so every coset has exactly $|H|$ elements. There are $[G : H]$ cosets. Summing their equal cardinalities gives $|G| = [G : H]|H|$.

Proof (Group-action proof)

Let G act transitively on the set G/H of left cosets by left multiplication. The stabilizer of the coset H is H . The orbit--stabilizer theorem therefore gives

$$|G| = |G \cdot H| |G_H| = |G/H| |H| = [G : H] |H|.$$

This argument uses orbit--stabilizer, whose usual proof already contains the same coset-counting idea.

PROBLEM • math:algebra:pb:z12-subgroups

Subgroups of the cyclic group of order 12

Work in the additive group $\mathbb{Z}/12\mathbb{Z}$.

List every subgroup of $\mathbb{Z}/12\mathbb{Z}$. For each subgroup, give its order and a generator. Explain why your list is complete.

Solution

A cyclic group has exactly one subgroup for each positive divisor of its order. The divisors of 12 are 1, 2, 3, 4, 6, 12, giving

order	subgroup	one generator
1	$\{0\}$	0
2	$\{0, 6\}$	6
3	$\{0, 4, 8\}$	4
4	$\{0, 3, 6, 9\}$	3
6	$\{0, 2, 4, 6, 8, 10\}$	2
12	$\mathbb{Z}/12\mathbb{Z}$	1

The classification theorem for subgroups of a cyclic group proves completeness.

Solution (Generator calculation)

For $k \in \mathbb{Z}/12\mathbb{Z}$, the element k has order

$$\frac{12}{\gcd(12, k)}.$$

Taking $k = 0, 6, 4, 3, 2, 1$ produces orders 1, 2, 3, 4, 6, 12, respectively. Each generated subgroup is the set of multiples of its generator modulo 12.